

Math 221 Homework 12

Atharv Sampath

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1 Question 7.1

Proof. Let $A \in \mathbb{R}^{n \times n}$ and let Q be orthogonal, so $Q^T Q = Q Q^T = I$. Set

$$\tilde{A} = Q^T A Q, \quad \tilde{q}_1 = Q^T q_1,$$

where $q_1 = q/\|q\|_2$ is the normalized starting vector for Arnoldi or Lanczos on A . Since Q is orthogonal, $\|\tilde{q}_1\|_2 = \|q_1\|_2 = 1$, so $\tilde{q}_1$ is a valid starting vector for the algorithms applied to $\tilde{A}$.

First consider the Arnoldi algorithm. Let q_j and h_{ij} be the basis vectors and Hessenberg entries produced when Arnoldi is run on (A, q_1) , and let $\tilde{q}_j, \tilde{h}_{ij}$ denote the quantities produced for $(\tilde{A}, \tilde{q}_1)$. Assume inductively that $\tilde{q}_i = Q^T q_i$ for $1 \leq i \leq j$. In the run on A the raw vector is $z = A q_j$, while in the run on $\tilde{A}$ it is

$$\tilde{z} = \tilde{A} \tilde{q}_j = Q^T A Q Q^T q_j = Q^T A q_j = Q^T z.$$

The inner products that define the Hessenberg entries satisfy

$$\tilde{h}_{ij} = \tilde{q}_i^T \tilde{z} = (Q^T q_i)^T (Q^T z) = q_i^T Q Q^T z = q_i^T z = h_{ij},$$

so the same coefficients h_{ij} are computed. The orthogonalization step on A forms

$$z' = z - \sum_{i=1}^j h_{ij} q_i,$$

while on $\tilde{A}$ we obtain

$$\tilde{z}' = \tilde{z} - \sum_{i=1}^j \tilde{h}_{ij} \tilde{q}_i = Q^T z - \sum_{i=1}^j h_{ij} Q^T q_i = Q^T \left(z - \sum_{i=1}^j h_{ij} q_i \right) = Q^T z'.$$

Because Q is orthogonal, $\|\tilde{z}'\|_2 = \|z'\|_2$, so $h_{j+1,j} = \|z'\|_2$ and $\tilde{h}_{j+1,j} = \|\tilde{z}'\|_2$ are equal. When $h_{j+1,j} \neq 0$ we get

$$\tilde{q}_{j+1} = \frac{\tilde{z}'}{\tilde{h}_{j+1,j}} = \frac{Q^T z'}{h_{j+1,j}} = Q^T q_{j+1}.$$

This completes the induction, so $\tilde{q}_i = Q^T q_i$ for all i and $\tilde{h}_{ij} = h_{ij}$ for all i, j . Hence the Hessenberg matrices H_k computed from (A, q) and $(Q^T A Q, Q^T q)$ are identical.

For the Lanczos algorithm we argue similarly. Let q_j, α_j, β_j be the vectors and scalars produced from (A, q_1) , and $\tilde{q}_j, \tilde{\alpha}_j, \tilde{\beta}_j$ those from $(\tilde{A}, \tilde{q}_1)$. Assume inductively that $\tilde{q}_i = Q^T q_i$ for $i \leq j$ and that $\tilde{\beta}_i = \beta_i$ for $i < j$. Again $\tilde{z} = Q^T z$ with $z = A q_j$, so

$$\tilde{\alpha}_j = \tilde{q}_j^T \tilde{z} = (Q^T q_j)^T (Q^T z) = q_j^T z = \alpha_j.$$

The residuals are

$$z' = z - \alpha_j q_j - \beta_{j-1} q_{j-1}, \quad \tilde{z}' = \tilde{z} - \tilde{\alpha}_j \tilde{q}_j - \tilde{\beta}_{j-1} \tilde{q}_{j-1} = Q^T z',$$

so $\tilde{\beta}_j = \|\tilde{z}'\|_2 = \|z'\|_2 = \beta_j$, and when $\beta_j \neq 0$,

$$\tilde{q}_{j+1} = \frac{\tilde{z}'}{\tilde{\beta}_j} = Q^T q_{j+1}.$$

Thus all α_j, β_j (and hence all tridiagonal matrices T_k) produced from (A, q) and $(Q^T A Q, Q^T q)$ are identical. This confirms that running Arnoldi or Lanczos on A with q yields the same H_k or T_k as running on $Q^T A Q$ with $Q^T q$. $\square$

2 Question 7.2

Proof. By Question 7.1 we may replace A by any orthogonally similar matrix without changing the tridiagonal matrices T_k produced by Lanczos. Thus we may assume that A is diagonal,

$$A = \text{diag}(\lambda_1, \dots, \lambda_n),$$

with orthonormal eigenvectors $e_1, \dots, e_n$. Let λ_i be simple with eigenvector e_i , and suppose $q_1 \perp e_i$. For diagonal A this is equivalent to the i th component of q_1 being zero.

For each $j \geq 0$, the vector $A^j q_1$ also has zero i th component, since the i th entry of $A^j q_1$ is $\lambda_i^j q_1(i) = 0$. Hence every Krylov subspace

$$\mathcal{K}_k(A, q_1) = \text{span}\{q_1, A q_1, \dots, A^{k-1} q_1\}$$

is contained in $e_i^\perp$. The Lanczos vectors $q_1, \dots, q_k$ are obtained by orthogonalizing these Krylov vectors, so

$$\text{span}\{q_1, \dots, q_k\} \subset e_i^\perp \quad \text{for all } k.$$

Run Lanczos in exact arithmetic until it stops at step m , i.e. until $\beta_m = 0$. Then we have the Lanczos decomposition

$$A Q_m = Q_m T_m, \quad Q_m^T Q_m = I_m,$$

so $S = \text{span}\{q_1, \dots, q_m\}$ is A -invariant and $T_m = Q_m^T A Q_m$ is the matrix of $A|_S$ in the orthonormal basis $\{q_j\}$. Since $S \subset e_i^\perp$ and λ_i is simple with eigenvector $e_i \notin S$, the restriction $A|_S$ cannot have eigenvalue λ_i : if $T_m y = \lambda_i y$ for some $y \neq 0$, then $x = Q_m y \in S$ would satisfy

$Ax = \lambda_i x$, providing an eigenvector for λ_i in S and contradicting the simplicity of λ_i and $x \perp e_i$. Thus $\lambda_i \notin \sigma(T_m)$.

Consequently, the eigenvalues of T_k cannot converge to λ_i in the sense that their limits, which are eigenvalues of the final tridiagonal T_m , equal λ_i . However, an intermediate T_k can still equal λ_i “accidentally”. For example, take

$$A = \text{diag}(3, 2, 1), \quad \lambda_2 = 2, \quad e_2 = (0, 1, 0)^T,$$

and choose $q_1 = (1, 0, 1)^T / \sqrt{2}$, which is orthogonal to e_2 . In the first Lanczos step,

$$z = Aq_1 = \frac{1}{\sqrt{2}}(3, 0, 1)^T, \quad \alpha_1 = q_1^T z = 2,$$

so $T_1 = [2]$ and its single eigenvalue equals λ_2 . Continuing,

$$z \leftarrow z - \alpha_1 q_1 = \frac{1}{\sqrt{2}}(1, 0, -1)^T, \quad \beta_1 = \|z\|_2 = 1, \quad q_2 = z/\beta_1 = \frac{1}{\sqrt{2}}(1, 0, -1)^T.$$

The second step yields

$$z = Aq_2 = \frac{1}{\sqrt{2}}(3, 0, -1)^T, \quad \alpha_2 = q_2^T z = 2,$$

and then

$$z \leftarrow z - \alpha_2 q_2 - \beta_1 q_1 = 0,$$

so $\beta_2 = 0$ and the algorithm stops with

$$T_2 = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix},$$

whose eigenvalues are 3 and 1, the eigenvalues of A associated with $e_2^\perp = \text{span}\{e_1, e_3\}$. The eigenvalue 2 is absent from T_2 , even though it appeared in T_1 . Thus the eigenvalues of the final tridiagonal T_m cannot converge to λ_i under the stated orthogonality condition, although intermediate T_k may equal λ_i by coincidence. $\square$

3 Question 7.3

Proof. In exact arithmetic, Lanczos applied to a symmetric A and starting vector q_1 produces orthonormal vectors $q_1, \dots, q_k$ and a symmetric tridiagonal matrix

$$T_k = \begin{bmatrix} \alpha_1 & \beta_1 & & \\ \beta_1 & \alpha_2 & \beta_2 & \\ & \ddots & \ddots & \ddots \\ & & \beta_{k-1} & \alpha_k \end{bmatrix},$$

where each $\beta_j = \|z\|_2 > 0$ as long as no breakdown occurs. Thus for every T_k that is actually formed, all subdiagonal entries $\beta_1, \dots, \beta_{k-1}$ are nonzero.

Fix such a k , and consider an eigenpair (θ, x) of T_k , where $x = (x_1, \dots, x_k)^T$. Writing $T_k x = \theta x$ componentwise yields the three-term recurrence

$$\beta_{j-1}x_{j-1} + \alpha_j x_j + \beta_j x_{j+1} = \theta x_j, \quad j = 1, \dots, k,$$

with $\beta_0 = \beta_k = 0$. From the first equation $(\alpha_1 - \theta)x_1 + \beta_1 x_2 = 0$, once x_1 is chosen, x_2 is uniquely determined because $\beta_1 \neq 0$. Then the equation for $j = 2$ determines x_3 uniquely from x_1, x_2 , and so on. By induction, all components of x are uniquely determined by the choice of x_1 , so the eigenspace corresponding to θ is at most one-dimensional. Since T_k is symmetric, its algebraic and geometric multiplicities coincide, and therefore every eigenvalue of T_k is simple. In particular, no symmetric tridiagonal matrix T_k produced by Lanczos can have an exactly multiple eigenvalue.

Now assume that $A \in \mathbb{R}^{n \times n}$ is symmetric and has a multiple eigenvalue. Suppose, for contradiction, that Lanczos applied to A runs to completion without breakdown, so that it produces n orthonormal vectors $q_1, \dots, q_n$ and an $n \times n$ tridiagonal matrix T_n . In exact arithmetic we then have

$$AQ_n = Q_n T_n, \quad Q_n^T Q_n = I_n,$$

so A and T_n are orthogonally similar and have the same eigenvalues with the same multiplicities. Because A has a multiple eigenvalue, T_n must also have a multiple eigenvalue, contradicting the fact that every eigenvalue of a Lanczos tridiagonal is simple. Hence the assumption that Lanczos does not break down before the last step is impossible.

Therefore, if A has a multiple eigenvalue, the Lanczos algorithm must encounter some $\beta_j = 0$ with $j < n$, and so it must break down before the last step. $\square$